of a mistake. The same is true for $|l\rangle$ and $|r\rangle$. You can distinguish them by measuring σ_x.

But suppose instead that you are told the spin is in one of the two states, $|u\rangle$ or $|r\rangle$ (*up* or *right*). There is nothing you can measure that will unambiguously tell you the spin's true state. Measuring σ_z won't do it. If you get $\sigma_z = +1$, it is possible that the initial state was $|r\rangle$ since there is a 50 percent probability of getting this answer in the state $|r\rangle$. For that reason, $|u\rangle$ and $|d\rangle$ are said to be physically distinguishable, but $|u\rangle$ and $|r\rangle$ are not. One might say that the inner product of two states is a measure of the inability to distinguish them with certainty. Sometimes this inner product is called the *overlap*. Principle 3 requires physically distinct states to be represented by orthogonal state-vectors, that is, vectors with no overlap. Thus, for spin states, $\langle u|d\rangle = 0$ but $\langle u|r\rangle = \frac{1}{\sqrt{2}}$.

Finally, Principle 4 quantifies these ideas in a rule that expresses the probabilities for various outcomes of an experiment. If we assume that a system has been prepared in state $|A\rangle$, and subsequently the observable **L** is measured, then the outcome will be one of the eigenvalues λ_i of the operator **L**. But, in general, there is no way to tell for certain which of these values will be observed. There is only a probability—let us call it $P(\lambda_i)$—that the outcome will be λ_i. Principle 4 tells us how to calculate that probability, and it is expressed in terms of the overlap of $|A\rangle$ and $|\lambda_i\rangle$. More precisely, the probability is the square of the magnitude of the overlap:

$$P(\lambda_i) = |\langle A|\lambda_i\rangle|^2$$

or, equivalently,

$$P(\lambda_i) = \langle A|\lambda_i\rangle\langle\lambda_i|A\rangle.$$

You might be wondering why the probability is not the overlap itself. Why the square of the overlap? Keep in mind that the inner product of two vectors is not always positive, or even real. Probabilities, on the other hand, are both